Lecture 18:
Variational Inference

October 26, 2026

Part IV: Simulation-Based Inference

return to Bayesian inference — now with ML

our **posterior sampling** arsenal:

rejection sampling

importance sampling

Markov Chain Monte Carlo (MCMC)

Hamiltonian Monte Carlo

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Hamiltonian Monte Carlo

requires differentiable posterior

variational inference

basic idea: pick an approximation $q_\phi(x)$ and then try to make it as close to the true posterior $p^*(x)$ by optimizing ϕ

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speed benefits of optimization + statistical benefits of the Bayesian approach

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can we quantify how closely $q(x)$ approximates $p^*(x)$?

information theory

information theory — information as **surprise**

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information from an event with probability p

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certain event: $p = 1 \rightarrow I = 0$ — *no surprise*

coin toss: $p = 0.5 \rightarrow I = 1$

rare event: $p \ll 1 \rightarrow I \gg 1$ — *surprising*

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"Nobody knows what entropy really is, so in a debate you will always have the advantage" — Von Neumann

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Observer B: believes world runs according to $q(x)$

The events are identical but B is more surprised than A on average, because their mental model is miscalibrated.

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$$H(p,q) = H(p) \iff p(x) = q(x)$$

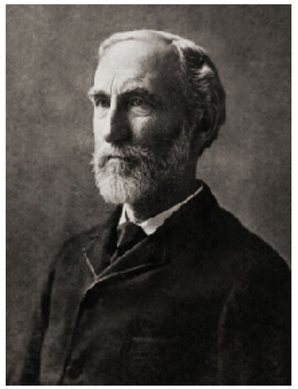
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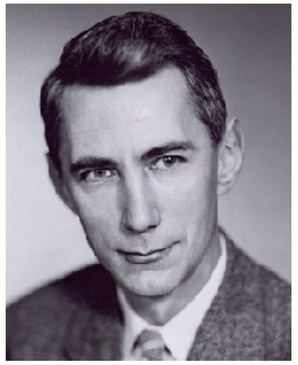
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Josiah Willard Gibbs



Claude Shannon

introduced by Gibbs in the 19th century for statistical mechanics applied to information theory by Shannon in 1948

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$$H(p, q) - H(p)$$

cross entropy — entropy

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"On Information and Sufficiency", Solomon Kullback and Richard A. Leibler *Annals of Mathematical Statistics* (1951)

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metric for measuring the dissimilarity of two distributions

caution: KL divergence is *not* a distance — $D_{\mathrm{KL}}(p \parallel q) \neq D_{\mathrm{KL}}(q \parallel p)$

back to variational inference — with KL divergence

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note the order of q_ϕ and p — expectation value with respect to q_ϕ, which we can choose to be easy to sample

$$= \left\langle \log \frac{q_\phi(x)}{p(x)} \right\rangle_{q_\phi}$$

variational inference: pick a surrogate distribution $q_\phi(x)$ then optimize $\min_\phi D_{\mathrm{KL}}\left(q_\phi(x) \parallel p^*(x)\right)$ to approximate $p^*(x)$

variational inference for **Bayesian inference**

target distribution is the posterior $p(\theta \mid X)$

variational inference for **Bayesian inference**

target distribution is the posterior $p(\theta \mid X)$ or $\hat{p} = p(X \mid \theta)\, p(\theta)$

unnormalized posterior

$\hat{p} = p\ Z$ $\qquad$ $Z = p(X)$ *is the* ***evidence***

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$$D_{\mathrm{KL}}(q_\phi \parallel \hat{p}) \geq - \log Z \quad \text{by Gibbs' inequality}$$

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minimizing KL is often called
Evidence Lower Bound (**ELBO**)

variational inference for Bayesian inference in practice

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$$\min_{\phi} D_{\mathrm{KL}}(q_\phi \parallel \hat{p}) = \int \mathrm{d}\theta \; q_\phi(\theta) \; \log \frac{q_\phi(\theta)}{\hat{p}(\theta \mid X)}$$

variational inference for Bayesian inference in practice

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Monte Carlo integration using $\theta' \sim q_\phi(\theta)$

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$$\approx \sum_{\theta' \sim q_\phi(\theta)} \left[\log q_\phi(\theta') - \log \hat{p}(\theta' \mid X)\right]$$

an approximate of the posterior $p(\theta \mid X) \approx q_\phi(\theta)$ that is easy to sample and evaluate

in-class: variational inference demo and in-class exercise

ast384c/inclass/lec18_vi/vi_exercise.ipynb